

Project Title:**AZURE LOAD BALANCER OUTBOUND RULES AND SNAT PLANNING****Abstract**

Cloud applications interact with many databases, external APIs, and many other cloud services. While interacting with these many services, they create many outbound connections. For maintaining the outbound traffic, cloud applications usually have load balancers, which send the outbound traffic through the Source Network Address Translation (SNAT) ports. But if the outbound connections increase too heavily, the SNAT ports in Azure application may become insufficient and may cause many application level problems.

The project mainly focuses on Azure Load Balancer, its outbound rules and how the SNAT ports are being planned to control the outbound traffic. For example, if we take any chatty application like an online shopping website, we have to study and analyse the outbound traffic and allocate the SNAT ports for different connections. We will design this application using Azure Load balancer, virtual machines and the required outbound connectivity. Then for testing purpose, we will send the sample outbound traffic through these SNAT ports and these SNAT ports have to analyse the outbound rules and we have to observe how well the SNAT ports are being able to control this traffic. Observe the SNAT port exhaustion and the traffic that each SNAT port is being able to handle.

In this project, it should observe and estimate the outbound traffic capacity for the SNAT ports, so that the port exhaustion can be reduced. For this, the SNAT ports will be tested by sending different configurations of outbound traffic through them. After analysing these configurations of traffic, the SNAT ports will be able to estimate the traffic and should maintain the ports such that the port exhaustion should be reduced.

At the end, this project's outcome should provide a practical understanding of Azure Load Balancer, outbound rules and how the SNAT ports are going to handle these. And also it shows the proper SNAT planning can improve the scalability of the cloud applications.